

CSE563 — Assignment 2

Unit II: Vectors and Matrices

Maps to CO2 (demonstrate the use of vectors and matrices in image transformations). **Marks: 50.** Attempt all questions. Show every step; state matrices in full.

Part A — Short Answer and Definitions (10 marks, 2 each)

Q1. [2 marks] Define linear independence. Determine whether $\mathbf{u} = (1, 2, 3)$, $\mathbf{v} = (2, 4, 6)$, $\mathbf{w} = (1, 0, 1)$ are linearly independent.

Q2. [2 marks] State the condition for a matrix A to be invertible, and give two equivalent characterisations. Is $A = \begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$ invertible?

Q3. [2 marks] Explain why a covariance matrix always has real eigenvalues and mutually orthogonal eigenvectors, and why this matters for PCA.

Q4. [2 marks] Why must translation be expressed in homogeneous coordinates rather than as a plain 2×2 matrix?

Q5. [2 marks] State one situation where nearest-neighbour interpolation is preferable to bilinear, and explain why.

Part B — Numerical Problems (26 marks)

Q6. *[4 marks]* For $\mathbf{u} = (2, -1, 2)$ and $\mathbf{v} = (1, 2, 2)$ compute: (a) $\mathbf{u} \cdot \mathbf{v}$; (b) $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$; (c) the angle between them; (d) the projection of $\mathbf{v}$ onto $\mathbf{u}$; (e) a unit vector along $\mathbf{u}$.

Q7. *[5 marks]* Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix}$. Compute (a) AB and BA , commenting on the result; (b) $\det A$; (c) A^{-1} , verifying $AA^{-1} = I$; (d) solve $A\mathbf{x} = (5, 11)^T$.

Q8. *[5 marks]* Find the eigenvalues and normalised eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$. Verify your eigenvalues using the trace and determinant, and confirm that the eigenvectors are orthogonal.

Q9. [6 marks] Perform PCA by hand on the five 2-D points $(2, 1), (3, 3), (4, 3), (5, 5), (6, 8)$. (a) Centre the data. (b) Compute the 2×2 covariance matrix. (c) Find both eigenvalues and the first principal component. (d) State the percentage of variance explained by PC1. (e) Project the point $(6, 8)$ onto PC1.

Q10. [6 marks] Two classes of 2-D samples: $C_1 = \{(2, 2), (3, 3), (4, 2), (3, 2)\}$ and $C_2 = \{(7, 6), (8, 7), (9, 6), (8, 6)\}$. (a) Compute the class means. (b) Compute S_1 , S_2 and S_W . (c) Find the LDA direction $\mathbf{w} \propto S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$. (d) Compute the threshold and classify the new point $(6, 5)$.

Part C — Image Transformation and Interpolation (14 marks)

Q11. [6 marks] A triangle has vertices $P_1 = (1, 1)$, $P_2 = (3, 1)$, $P_3 = (1, 4)$. (a) Write the 3×3 homogeneous matrices for a scaling by $(2, 2)$ and a translation by $(3, -1)$. (b) Give the single matrix M that scales first and then translates, and apply it to all three vertices. (c) Give the matrix for the opposite order and apply it to P_1 only. Comment on the difference.

Q12. *[4 marks]* Rotate the point $(4, 2)$ by 90° anticlockwise about the point $(2, 2)$. Write the composite matrix and verify your result geometrically.

Q13. *[4 marks]* A grayscale image has pixel values $I(0, 0) = 100$, $I(1, 0) = 150$, $I(0, 1) = 200$, $I(1, 1) = 50$ (first index = x /column, second = y /row). Estimate the intensity at $(0.4, 0.7)$ by (a) nearest-neighbour and (b) bilinear interpolation. Show the four weights and verify they sum to 1.